

Functions



Evaluating functions

- 1 If $f(x) = 3x - 4$, find $f(5)$.
 - 2 If $f(x) = x^2 - 2x + 1$, find $f(-2)$.
 - 3 If $g(x) = \sqrt{x + 5}$, find $g(4)$.
 - 4 If $h(x) = \frac{2x}{x - 1}$, find $h(3)$.
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Domain and range

- 5 State the domain of $f(x) = \sqrt{x - 3}$.
 - 6 State the domain of $f(x) = \frac{1}{x - 5}$.
 - 7 State the domain of $f(x) = \sqrt{7 - x}$.
 - 8 State the range of $f(x) = x^2 + 2$.
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Transformations: shifts

- 9 The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.
 - 10 The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.
 - 11 Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.
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Transformations: reflections and stretches

- 12 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 3. Write the new equation.
- 13 Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.
- 14 Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.

Composite functions

- 15 Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.
- 16 Using the same f and g , find $(g \circ f)(2)$.
- 17 Using the same f and g , find a general formula for $(f \circ g)(x)$.
- 18 Using the same f and g , find a general formula for $(g \circ f)(x)$.
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Inverse functions

- 19 Find the inverse of $f(x) = 3x - 2$.
- 20 Find the inverse of $f(x) = \frac{x - 4}{5}$.
- 21 Show that $f(x) = 2x + 6$ and $g(x) = \frac{x - 6}{2}$ are inverses by computing $f(g(x))$.
- 22 Find the inverse of $f(x) = x^3 + 1$.
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Piecewise functions

- 23 A function is defined by $f(x) = x + 2$ for $x < 1$, and $f(x) = 3x - 1$ for $x \geq 1$. Find $f(-2)$.
- 24 Using the same piecewise function, find $f(1)$.
- 25 Using the same piecewise function, find $f(4)$.
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Even and odd functions

- 26 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.
- 27 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.
- 28 Determine whether $f(x) = x^2 + x$ is even, odd, or neither.
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Increasing, decreasing, and average rate of change

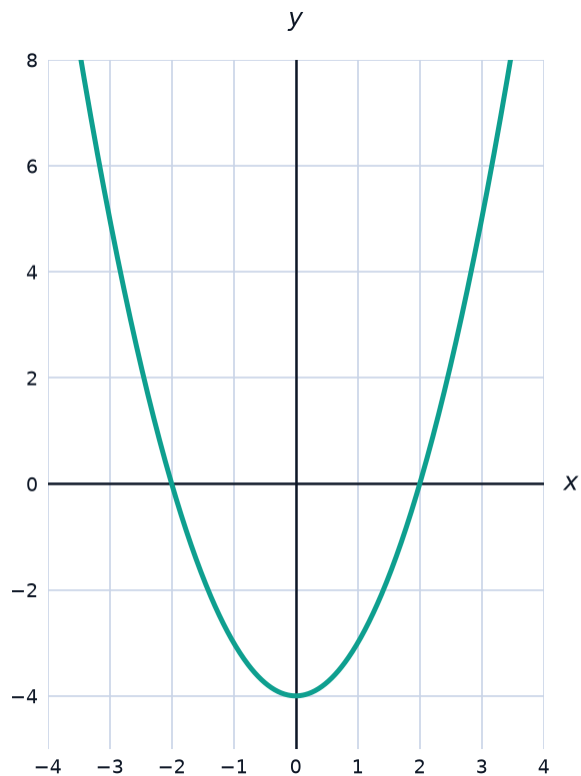
- 29 Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.
- 30 Find the average rate of change of $f(x) = 2x^2 - 3x$ from $x = 0$ to $x = 3$.
- 31 State the interval on which $f(x) = (x - 2)^2$ is decreasing.
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Operations on functions

- 32 Let $f(x) = x + 3$ and $g(x) = x^2 - 1$. Find $(f + g)(x)$.
- 33 Using the same f and g , find $(f \cdot g)(2)$.
- 34 Using the same f and g , find $(f/g)(3)$.
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Reading a function graph

- 35 The graph of $y = f(x) = x^2 - 4$ is shown.



- a State the zeros of f .
- b State the domain and range of f .

Word problems: functions in context

- 36** This question has two parts. A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 150$.
- a** Evaluate $P(10)$ to find the profit from selling 10 units.
 - b** Find the number of units x that maximizes profit (using the vertex of the parabola), and state the maximum profit.
- 37** This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city t hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \leq t \leq 12$.
- a** Find $C(4)$, the Celsius temperature 4 hours after 6 a.m.
 - b** Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.

MCQ — Evaluating functions

- 38** If $f(x) = 4x - 7$, find $f(6)$.
- a.** 17
 - b.** 24
 - c.** 31
 - d.** 10
- 39** If $f(x) = x^2 - 3x + 2$, find $f(-3)$.
- a.** 2
 - b.** -16
 - c.** 20
 - d.** 16

40 If $g(x) = \sqrt{x + 9}$, find $g(7)$.

- a. 8
- b. 16
- c. 4
- d. $\sqrt{16}$

41 If $h(x) = \frac{3x}{x-2}$, find $h(5)$.

- a. 15
- b. 5
- c. 3
- d. 1.5

42 If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

- a. 18
- b. 8
- c. 16
- d. 22

MCQ — Domain and range

43 State the domain of $f(x) = \sqrt{x - 5}$.

- a. $x > 5$
- b. $x \geq 5$
- c. $x \leq 5$
- d. $x \neq 5$

44 State the domain of $f(x) = \frac{1}{x-7}$.

a. All reals except $x = 7$

b. $x > 7$

c. $x \geq 7$

d. All real numbers

45 State the domain of $f(x) = \sqrt{9-x}$.

a. $x \leq 9$

b. $x < 9$

c. $x \neq 9$

d. $x \geq 9$

46 State the range of $f(x) = x^2 + 5$.

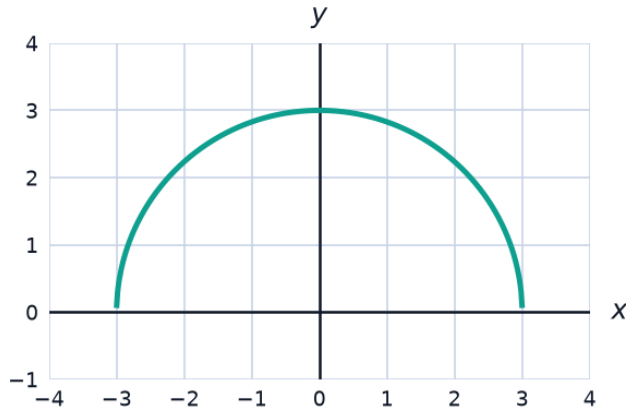
a. $y > 5$

b. All real numbers

c. $y \leq 5$

d. $y \geq 5$

- 47 The graph shows $f(x)$ defined only where plotted. State its domain.



- a. $[-3, 3]$
- b. All real numbers
- c. $[-4, 4]$
- d. $[0, 3]$

MCQ — Transformations: shifts

- 48 The graph of $y = x^2$ is shifted 5 units right and 3 units down. Write the new equation.

- a. $y = (x - 5)^2 + 3$
- b. $y = (x - 5)^2 - 3$
- c. $y = (x + 5)^2 - 3$
- d. $y = (x - 3)^2 - 5$

- 49 The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

- a. $y = |x + 4| + 2$
- b. $y = |x + 2| + 4$
- c. $y = |x + 2| - 4$
- d. $y = |x - 2| + 4$

50 Describe the transformation that takes $y = x^2$ to $y = (x - 3)^2 + 2$.

- a. Shift right 2, up 3
- b. Shift right 3, down 2
- c. Shift left 3, up 2
- d. Shift right 3, up 2

51 The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

- a. $y = \sqrt{x + 6}$
- b. $y = \sqrt{x} - 6$
- c. $y = \sqrt{x} + 6$
- d. $y = \sqrt{x - 6}$

MCQ — Transformations: reflections and stretches

52 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 2. Write the new equation.

- a. $y = -\frac{1}{2}x^2$
- b. $y = (-2x)^2$
- c. $y = -2x^2$
- d. $y = 2x^2$

53 Describe the transformation that takes $y = \sqrt{x}$ to $y = 3\sqrt{x}$.

- a. Reflection over the x -axis
- b. Vertical shift up 3
- c. Vertical stretch by a factor of 3
- d. Horizontal stretch by a factor of 3

- 54 Describe the transformation that takes $y = x^2$ to $y = -(x + 2)^2 + 1$.
- a. Reflect over x -axis, shift right 2, down 1
 - b. Reflect over y -axis, shift right 2, up 1
 - c. Reflect over x -axis, shift left 2, up 1
 - d. Shift left 2, up 1 (no reflection)
- 55 The graph of $y = \sqrt{x}$ is reflected over the y -axis. Write the new equation.
- a. $y = -\sqrt{-x}$
 - b. $y = -\sqrt{x}$
 - c. $y = \sqrt{-x}$
 - d. $y = \sqrt{x}$
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MCQ — Composite functions

- 56 Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. Find $(f \circ g)(3)$.
- a. 26
 - b. 23
 - c. 11
 - d. 29
- 57 Using the same f and g , find $(g \circ f)(3)$.
- a. 100
 - b. 120
 - c. 121
 - d. 26

58 Using the same f and g , find a general formula for $(f \circ g)(x)$.

a. $3x^2 - 1$

b. $9x^2 - 1$

c. $3x^2 - 3$

d. $3x^2 + 2$

59 Using the same f and g , find a general formula for $(g \circ f)(x)$.

a. $9x^2 + 12x + 3$

b. $3x^2 + 2x - 1$

c. $9x^2 + 4$

d. $9x^2 + 12x - 1$

60 Let $f(x) = x + 4$ and $g(x) = 2x$. Find $(f \circ g)(5)$.

a. 10

b. 9

c. 14

d. 18

MCQ — Inverse functions

61 Find the inverse of $f(x) = 4x - 3$.

a. $f^{-1}(x) = \frac{x-3}{4}$

b. $f^{-1}(x) = \frac{x+4}{3}$

c. $f^{-1}(x) = \frac{x+3}{4}$

d. $f^{-1}(x) = 4x + 3$

- 62 Find the inverse of $f(x) = \frac{x-7}{2}$.
- a. $f^{-1}(x) = 2x - 7$
 - b. $f^{-1}(x) = 2(x - 7)$
 - c. $f^{-1}(x) = 2x + 7$
 - d. $f^{-1}(x) = \frac{x+7}{2}$
- 63 Show that $f(x) = 3x + 9$ and $g(x) = \frac{x-9}{3}$ are inverses by computing $f(g(x))$.
- a. x
 - b. $x + 9$
 - c. $3x$
 - d. $x - 9$
- 64 Find the inverse of $f(x) = x^3 - 2$.
- a. $f^{-1}(x) = \sqrt[3]{x} + 2$
 - b. $f^{-1}(x) = (x + 2)^3$
 - c. $f^{-1}(x) = \sqrt[3]{x - 2}$
 - d. $f^{-1}(x) = \sqrt[3]{x + 2}$
- 65 Find the inverse of $f(x) = 5x + 10$.
- a. $f^{-1}(x) = \frac{x-10}{5}$
 - b. $f^{-1}(x) = \frac{x+10}{5}$
 - c. $f^{-1}(x) = \frac{x}{5} - 10$
 - d. $f^{-1}(x) = 5x - 10$

MCQ — Piecewise functions

- 66 A function is defined by $f(x) = x + 3$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $f(-3)$.
- a. -7
 - b. 0
 - c. 6
 - d. 1
- 67 Using the same piecewise function, find $f(2)$.
- a. 3
 - b. 4
 - c. 0
 - d. 5
- 68 Using the same piecewise function, find $f(5)$.
- a. 5
 - b. 9
 - c. 7
 - d. 8
- 69 A function is defined by $f(x) = 2x$ for $x \leq 0$, and $f(x) = x^2 + 1$ for $x > 0$. Find $f(0) + f(3)$.
- a. 1
 - b. 11
 - c. 0
 - d. 10

MCQ — Even and odd functions

70 Determine whether $f(x) = x^2 - 7$ is even, odd, or neither.

- a. Neither
- b. Even
- c. Odd
- d. Both even and odd

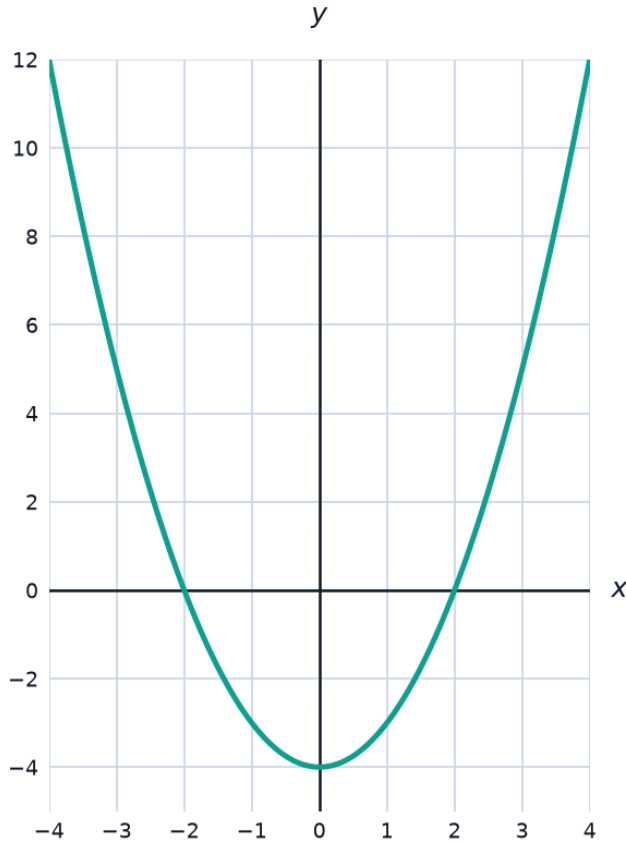
71 Determine whether $f(x) = x^3 - 4x$ is even, odd, or neither.

- a. Neither
- b. Both even and odd
- c. Even
- d. Odd

72 Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

- a. Both even and odd
- b. Neither
- c. Odd
- d. Even

73 The graph shown is symmetric about the y -axis. Is the function even, odd, or neither?



- a. Odd
- b. Neither
- c. Cannot be determined
- d. Even

MCQ — Increasing, decreasing, and average rate of change

74 Find the average rate of change of $f(x) = x^2$ from $x = 2$ to $x = 5$.

- a. 3
- b. 7
- c. 21
- d. 29

75 Find the average rate of change of $f(x) = 3x^2 - 2x$ from $x = 0$ to $x = 4$.

a. 40

b. 8.5

c. 10

d. 46

76 State the interval on which $f(x) = (x - 3)^2$ is decreasing.

a. $(-\infty, 3)$

b. $(-\infty, -3)$

c. $(-3, \infty)$

d. $(3, \infty)$

77 Find the average rate of change of $f(x) = x^3$ from $x = 1$ to $x = 3$.

a. 8

b. 26

c. 9

d. 13

MCQ — Operations on functions

78 Let $f(x) = x + 5$ and $g(x) = x^2 - 2$. Find $(f + g)(x)$.

a. $x^2 - x + 3$

b. $x^2 + x + 3$

c. $x^2 + x + 7$

d. $x^2 + 2x + 3$

79 Using the same f and g , find $(f \cdot g)(3)$.

a. 56

b. 48

c. 62

d. 50

80 Using the same f and g , find $(f/g)(4)$.

a. $\frac{7}{14}$

b. $\frac{9}{16}$

c. $\frac{9}{14}$

d. $\frac{14}{9}$

81 Let $f(x) = 2x$ and $g(x) = x + 6$. Find $(f - g)(x)$.

a. $3x + 6$

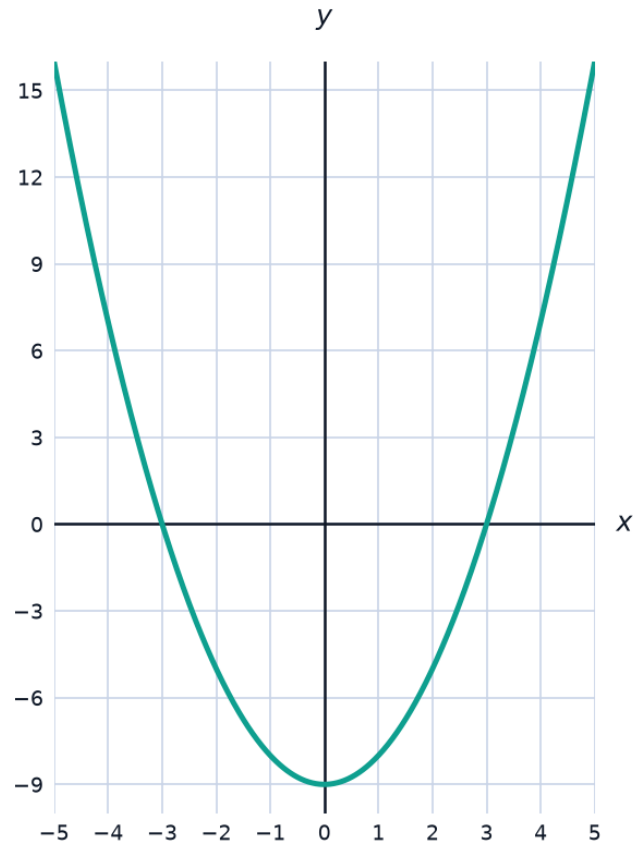
b. $x - 6$

c. $x + 6$

d. $-x - 6$

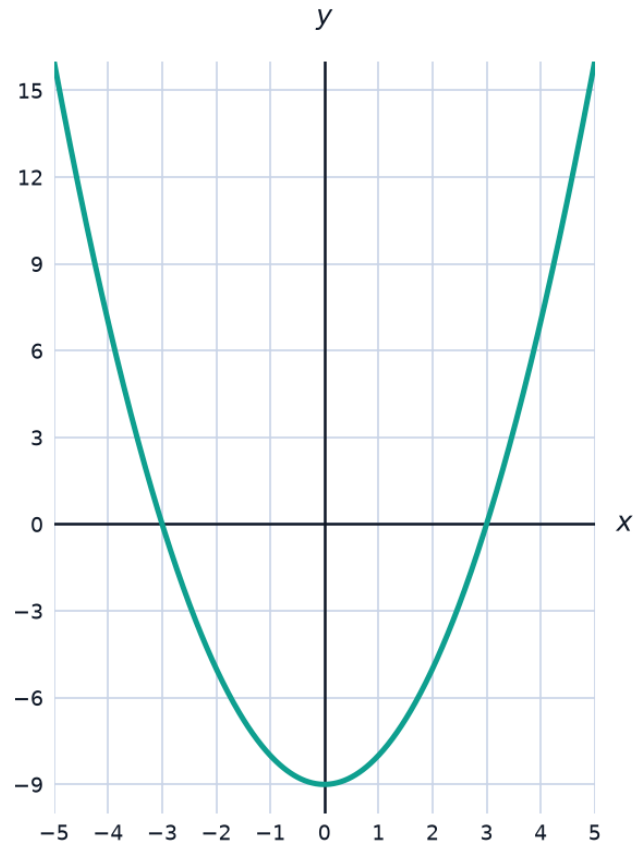
MCQ — Reading a function graph

- 82 The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.



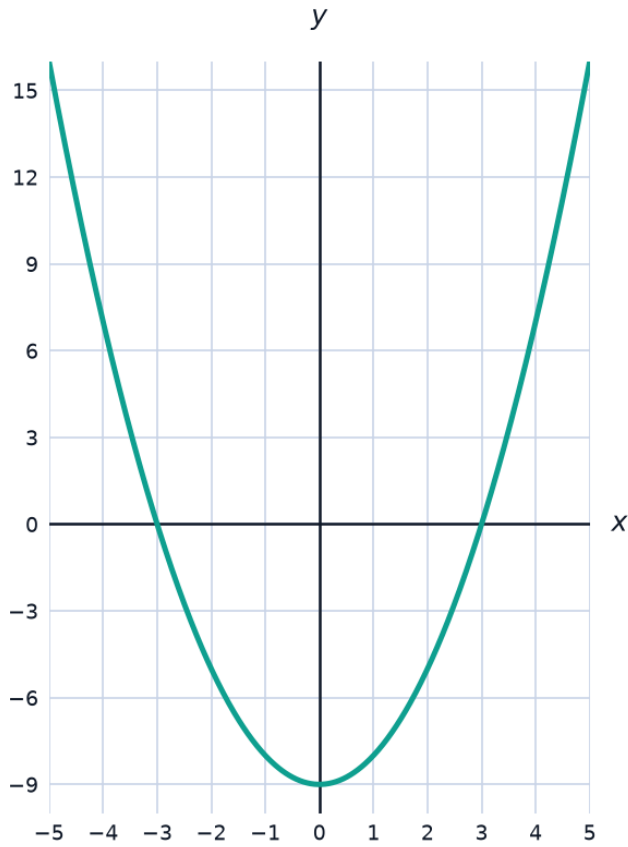
- a. -9
- b. 6
- c. 9
- d. 0

- 83 Using the same graph, find the values of x where $f(x) = 0$ (the x -intercepts).



- a. $x = 0$ and $x = 9$
- b. $x = -9$ and $x = 9$
- c. $x = 3$ only
- d. $x = -3$ and $x = 3$

84 Using the same graph, find the y -intercept.



- a. -3
- b. 9
- c. 0
- d. -9

MCQ — Word problems: functions in context

- 85 A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.
- a. 450 dollars
 - b. 150 dollars
 - c. 600 dollars
 - d. 900 dollars
- 86 The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.
- a. 72°F
 - b. 68°F
 - c. 36°F
 - d. 52°F
- 87 A ball's height (in meters) t seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.
- a. 2
 - b. 18
 - c. 22
 - d. 42

Multiple choice — easy

- 88 If $f(x) = x^2 - 6x - 7$, find $f(6)$.
- a. -7
 - b. 7
 - c. -10
 - d. -4

89 If $f(x) = -3x^2 - 5x + 9$, find $f(-3)$.

a. 0

b. -6

c. -3

d. 3

90 If $f(x) = -x^2 + 7x + 1$, find $f(-1)$.

a. -4

b. 7

c. -7

d. -10

91 If $f(x) = -3x^2 - x - 1$, find $f(4)$.

a. -56

b. 53

c. -50

d. -53

92 If $f(x) = x^2 - 5x - 6$, find $f(-3)$.

a. 21

b. 15

c. -18

d. 18

- 93 If $f(x) = x^2 + 8x - 1$, find $f(-5)$.
- a. -19
 - b. -16
 - c. -13
 - d. 16
- 94 If $f(x) = 2x^2 - 2x + 3$, find $f(-4)$.
- a. 40
 - b. 43
 - c. 46
 - d. -43
- 95 If $f(x) = 2x^2 + 8x + 5$, find $f(-1)$.
- a. -4
 - b. -1
 - c. 1
 - d. 2
- 96 If $f(x) = -3x^2 - 8x + 3$, find $f(-1)$.
- a. 8
 - b. 11
 - c. -8
 - d. 5

97 If $f(x) = 2x^2 + x + 8$, find $f(3)$.

a. -29

b. 32

c. 26

d. 29

98 If $f(x) = x^2 + 6x + 9$, find $f(-5)$.

a. 4

b. 1

c. 7

d. -4

99 If $f(x) = -3x^2 + 6x + 1$, find $f(-1)$.

a. -8

b. 8

c. -11

d. -5

100 If $f(x) = 3x^2 - 4$, find $f(-4)$.

a. 41

b. 47

c. -44

d. 44

101 If $f(x) = x^2 - 8x - 3$, find $f(5)$.

- a. -21
- b. 18
- c. -18
- d. -15

102 If $f(x) = x^2 - 2x - 2$, find $f(-2)$.

- a. 9
- b. -6
- c. 3
- d. 6

Multiple choice — medium

103 State the domain of $f(x) = \sqrt{-x - 6}$.

- a. $x \leq -6$
- b. $x \leq -4$
- c. $x \leq -8$
- d. $x \geq -6$

104 State the domain of $f(x) = \frac{1}{x - 9}$.

- a. $x \neq 7$
- b. $x = 9$
- c. $x \neq 9$
- d. $x \neq 11$

105 State the domain of $f(x) = \frac{1}{x-7}$.

- a. $x \neq 5$
- b. $x = 7$
- c. $x \neq 9$
- d. $x \neq 7$

106 State the domain of $f(x) = \frac{1}{x-11}$.

- a. $x \neq 9$
- b. $x \neq 13$
- c. $x = 11$
- d. $x \neq 11$

107 State the domain of $f(x) = \sqrt{2x-2}$.

- a. $x \geq -1$
- b. $x \geq 3$
- c. $x \leq 1$
- d. $x \geq 1$

108 State the domain of $f(x) = \sqrt{3x+6}$.

- a. $x \geq -2$
- b. $x \geq -4$
- c. $x \leq -2$
- d. $x \geq 0$

109 State the domain of $f(x) = \sqrt{-3x + 24}$.

a. $x \leq 8$

b. $x \leq 6$

c. $x \leq 10$

d. $x \geq 8$

110 State the domain of $f(x) = \sqrt{-2x + 4}$.

a. $x \leq 4$

b. $x \geq 2$

c. $x \leq 2$

d. $x \leq 0$

111 State the domain of $f(x) = \frac{1}{x+3}$.

a. $x \neq -1$

b. $x \neq -5$

c. $x \neq -3$

d. $x = -3$

112 State the domain of $f(x) = \sqrt{x - 2}$.

a. $x \leq 2$

b. $x \geq 4$

c. $x \geq 0$

d. $x \geq 2$

113 State the domain of $f(x) = \frac{1}{x+5}$.

a. $x \neq -5$

b. $x \neq -7$

c. $x = -5$

d. $x \neq -3$

114 State the domain of $f(x) = \frac{1}{x-1}$.

a. $x = 1$

b. $x \neq 1$

c. $x \neq -1$

d. $x \neq 3$

115 State the domain of $f(x) = \sqrt{-3x-9}$.

a. $x \leq -1$

b. $x \leq -3$

c. $x \geq -3$

d. $x \leq -5$

116 State the domain of $f(x) = \sqrt{2x-8}$.

a. $x \geq 2$

b. $x \geq 4$

c. $x \leq 4$

d. $x \geq 6$

117 State the domain of $f(x) = \sqrt{-2x + 14}$.

- a. $x \leq 7$
 - b. $x \leq 5$
 - c. $x \geq 7$
 - d. $x \leq 9$
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Multiple choice — hard

118 Let $f(x) = -x + 4$ and $g(x) = x^2 - 5$. Find $(f \circ g)(0)$.

- a. 6
- b. 9
- c. -9
- d. 12

119 Let $f(x) = 3x + 2$ and $g(x) = 2x^2 + 3$. Find $(f \circ g)(0)$.

- a. 8
- b. -11
- c. 11
- d. 14

120 Let $f(x) = -2x + 8$ and $g(x) = -x^2$. Find $(f \circ g)(4)$.

- a. 40
- b. -40
- c. 43
- d. 37

121 Let $f(x) = -x - 2$ and $g(x) = -x^2 + 3$. Find $(f \circ g)(1)$.

a. -4

b. -7

c. -1

d. 4

122 Let $f(x) = x - 6$ and $g(x) = 2x^2 - 3$. Find $(f \circ g)(2)$.

a. -4

b. -1

c. 1

d. 2

123 Find the inverse of $f(x) = 5x - 8$.

a. $\frac{x - 8}{5}$

b. $\frac{x + 8}{-5}$

c. $\frac{x + 8}{5}$

d. $5x - 8$

124 Find the inverse of $f(x) = 3x + 5$.

a. $3x + 5$

b. $\frac{x - 5}{3}$

c. $\frac{x + 5}{3}$

d. $\frac{x - 5}{-3}$

125 Find the inverse of $f(x) = 4x - 5$.

a. $\frac{x-5}{4}$

b. $\frac{x+5}{-4}$

c. $\frac{x+5}{4}$

d. $4x - 5$

126 Find the inverse of $f(x) = 4x - 3$.

a. $\frac{x-3}{4}$

b. $4x - 3$

c. $\frac{x+3}{-4}$

d. $\frac{x+3}{4}$

127 Find the inverse of $f(x) = 5x + 1$.

a. $\frac{x-1}{5}$

b. $\frac{x-1}{-5}$

c. $\frac{x+1}{5}$

d. $5x + 1$

128 The graph of $y = x^2$ is shifted 4 units right and 2 units down. Write the new equation.

a. $y = (x + 4)^2 - 2$

b. $y = (x + 4)^2 + 2$

c. $y = (x - 4)^2 - 2$

d. $y = (x - 4)^2 + 2$

129 The graph of $y = |x|$ is shifted 3 units right and 2 units down. Write the new equation.

a. $y = |x + 3| + 2$

b. $y = |x + 3| - 2$

c. $y = |x - 3| - 2$

d. $y = |x - 3| + 2$

130 The graph of $y = x^2$ is shifted 1 unit left and 4 units down. Write the new equation.

a. $y = (x - 1)^2 + 4$

b. $y = (x + 1)^2 + 4$

c. $y = (x + 1)^2 - 4$

d. $y = (x - 1)^2 - 4$

131 The graph of $y = |x|$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = |x + 7| + 3$

b. $y = |x - 7| - 3$

c. $y = |x + 7| - 3$

d. $y = |x - 7| + 3$

132 The graph of $y = x^2$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = (x - 7)^2 + 3$

b. $y = (x - 7)^2 - 3$

c. $y = (x + 7)^2 + 3$

d. $y = (x + 7)^2 - 3$